MECHATRONICS AND MEMS

Subject Code : 23A03702d

UNIT III : Actuators

Syllabus: Mechanical, Electrical, Hydraulic and Pneumatic Actuation systems, Characteristics and their limitations, Design of Hydraulic and Pneumatic circuits, Piezoelectric actuators, Shape memory alloys, Selection criteria for actuators.

I. Introduction to Actuators

- An **actuator** is the output device of a mechatronic system. It takes the low power control signal from the controller and converts it into **physical action** - linear or rotary motion, force, torque, heat - to drive the load.
- Actuator = **power converter** (electrical / fluid energy -> mechanical energy). The controller output is amplified by a **driver** (relay, transistor, SCR, servo amplifier, valve).
- **Types of actuation systems**
  - **Mechanical** (gears, cams, belts, linkages)
  - **Electrical** (solenoids, relays, DC / AC / stepper / servo motors)
  - **Hydraulic** (oil under high pressure)
  - **Pneumatic** (compressed air)
  - **Smart material actuators** (piezoelectric, shape memory alloy, magnetostrictive)

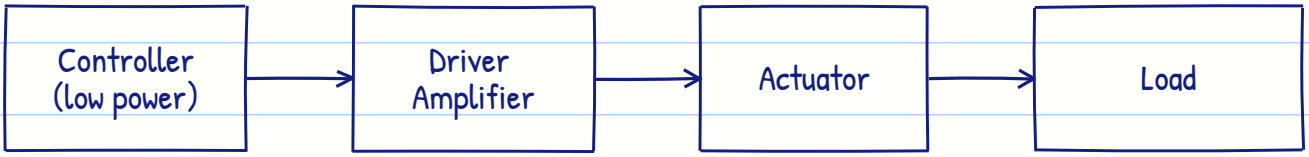


Fig: Actuator in a mechatronic system

2. Mechanical Actuation Systems

Mechanical systems transmit and change motion, speed, torque and direction between the prime mover and the load.

2.1 Gears and gear trains

- Change speed and torque. Gear ratio G = N1 / N2 = T2 / T1 = z2 / z1. A speed reducer increases torque.
- Types: spur, helical, bevel, worm (high reduction, self locking), rack and pinion (rotary to linear), planetary and harmonic drives (robots, compact, no backlash).
- **Backlash** is the play between meshing teeth - it causes positioning error; it is reduced by anti-backlash gears.

2.2 Belt and chain drives

- Flat, V, **timing (toothed) belts** and chains transmit power over long distance. Timing belt has no slip, so used in 3D printers and CNC.

2.3 Cams and followers

- A cam converts rotary motion into a precise, programmed reciprocating motion of the follower. Types: disc, cylindrical, translating cams. Used in engines (valve operation), packaging and automatic machines.

2.4 Ratchet and pawl, Geneva mechanism, linkages

- **Ratchet and pawl:** one directional motion; **Geneva wheel:** gives intermittent (indexing) rotation; **four bar linkage and slider crank:** convert rotation to oscillation or reciprocation.

2.5 Bearings, lead and ball screws

- **Lead screw / Ball screw and nut:** convert rotary motion to linear motion. Ball screw has recirculating balls so friction is very low and efficiency is 90%+; used in all CNC axes. Linear displacement = lead x revolutions.
- **Bearings** (ball, roller, linear guides) support the load and reduce friction.

Characteristics and limitations

- Reliable, simple and robust, give high torque. But they have **backlash, friction, wear** , need lubrication, are bulky, and the motion is fixed (not flexible) except for cams that need changing.

3. Electrical Actuation Systems

Electrical actuators are the most common - they are clean, easy to control, accurate, and compatible with electronics.

3.1 Switching devices

- **Mechanical switches and relays:** electromagnetic relay - a small current in the coil closes contacts of a heavy load circuit. Isolation, but slow and contacts wear.
- **Solid state relay (SSR), diode, thyristor (SCR), triac, BJT, MOSFET, IGBT:** electronic switches for fast, silent, long life switching and speed control of motors.

3.2 Solenoids

- A coil with a movable iron plunger. When current flows, the magnetic field pulls the plunger in (linear motion). Spring returns it back. Force is proportional to (N I)^2.
- Used in solenoid valves, door locks, relays, printers. Short stroke, on/off action only.

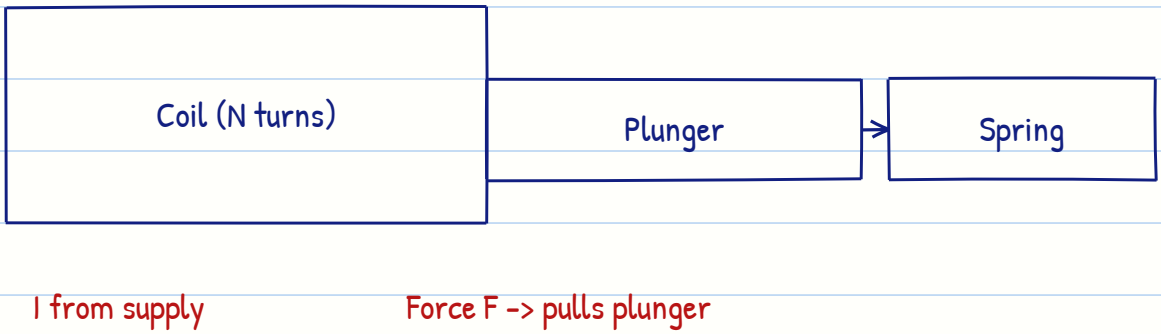


Fig: Solenoid actuator

3.3 DC motors

- Torque T = K phi Ia. Back emf Eb = K phi w. Speed w = (V - Ia Ra) / (K phi).
- **Permanent magnet DC (PMDC), separately excited, series and shunt** types. Speed control by armature voltage using **PWM** with H-bridge (also gives reversing).
- Advantages: good starting torque, easy speed control, linear torque-speed characteristic. **Limitations:** brushes and commutator wear, sparking, noise, maintenance.
- **Brushless DC motor (BLDC):** permanent magnet rotor, electronic commutation using hall sensors; no brush wear, high efficiency, long life, higher cost. Used in drones, EV, hard disks.

3.4 AC motors

- **Induction motor:** rugged, cheap, little maintenance; speed Ns = 120 f / P. Speed control using **VFD (variable frequency drive)** V/f control. Used in conveyors, pumps, machine spindles.
- **Synchronous motor:** runs at exact synchronous speed; used with servo drives (PMSM).

3.5 Stepper motors

- Rotate in fixed **steps** per pulse (1.8 deg per step = 200 steps/rev). Open loop position control; no feedback needed.
- Types: **permanent magnet, variable reluctance and hybrid** . Step angle = 360 / (number of steps). Speed (rpm) = (step angle x pulse rate x 60) / 360.
- Driven by a driver using full step, half step or micro stepping.

  Advantages: precise, cheap, holding torque, simple control. **Limitations:** loss of steps at high load or speed, vibration/resonance, low efficiency, torque falls with speed.
- Uses: 3D printers, CNC small machines, camera, printers, robots.

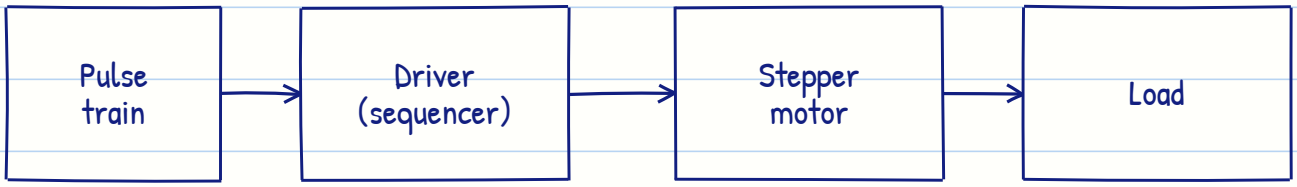


Fig: Stepper motor control (open loop)

3.6 Servo motors

- A **servo motor** is a motor + position/speed sensor (encoder or resolver) + controller in **closed loop** . It follows the command position exactly, with fast response and high torque at high speed.
- Types: DC servo and AC servo (PMSM / BLDC). Control: position loop, velocity loop and current loop (cascade PID).
- **Advantages:** high accuracy, high speed, good torque to inertia ratio, handles load changes. **Limitations:** costlier than stepper, needs tuning, feedback device needed, complex.

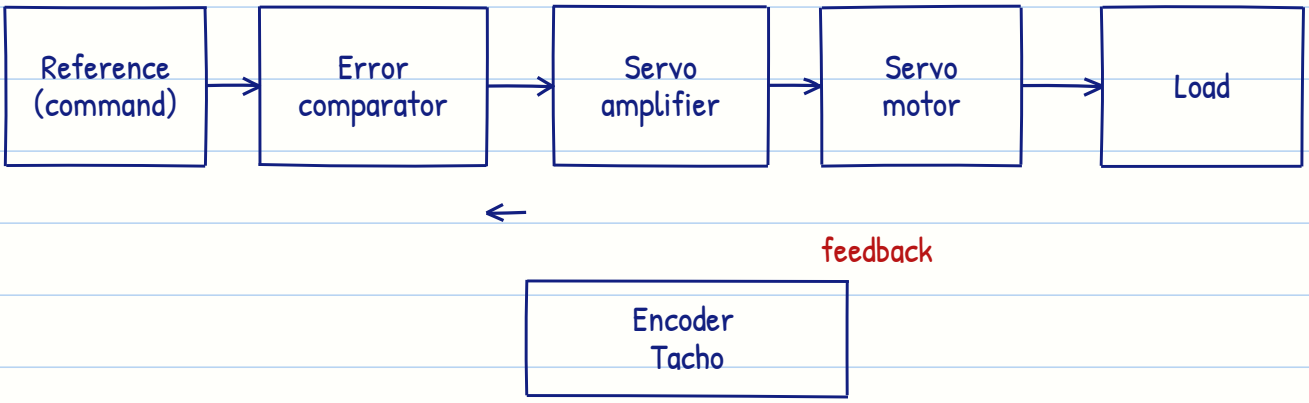


Fig: Closed loop servo system

Point	Stepper motor	Servo motor
Control	Open loop (no feedback)	Closed loop (encoder)
Cost	Low	High
Speed / torque	Torque falls fast with speed	Good torque at high speed
Accuracy	Good, may lose steps	Very high, no step loss
Use	3D printer, small CNC	CNC, robots, high performance

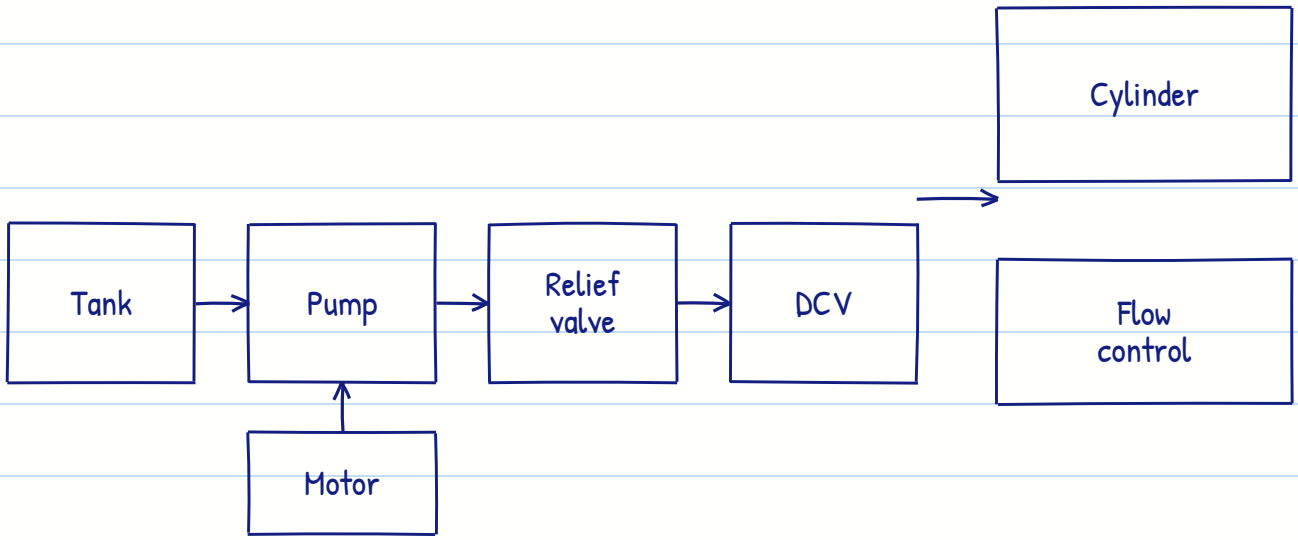
4. Hydraulic Actuation Systems

- Use **incompressible liquid (mineral oil)** at high pressure (up to 350 bar) to transmit power. Based on **Pascal's law:** pressure applied to a confined fluid is transmitted equally in all directions. $F = P \times A$.
- Gives **very large forces** from small size, smooth and stiff motion, good for heavy loads - presses, excavators, aircraft controls, machine tools, robots (large).

Components of a hydraulic system

- **Reservoir (tank):** stores oil, cools and settles dirt.
- **Pump:** converts mechanical energy to fluid energy. Gear, vane and **piston pumps** (positive displacement), driven by electric motor.
- **Pressure relief valve:** safety valve limits maximum pressure.
- **Filters and strainers,** coolers, accumulators and pressure gauges.
- **Control valves:** direction control valve (DCV), pressure control and flow control (needle) valves.
- **Actuators: cylinders** (linear: single acting, double acting) and **hydraulic motors** (rotary).
- Pipes, hoses and fittings.

Fig - Basic hydraulic power pack



Hydraulic cylinders

- **Single acting:** pressure on one side only, spring or load returns it. **Double acting:** pressure on both sides - force both ways. Force extend $F = P \times A$; retract $F = P \times (A - a)$ (rod area a). Speed $v = Q / A$.

Hydraulic directional control valves

- Described by **ways and positions** - eg: **4/3 valve** (4 ports, 3 positions) with centre positions closed, open or tandem. Operated by lever, solenoid, pilot pressure; **servo valves** and **proportional valves** give continuous control.

Advantages

- Large force and power to weight ratio, easy speed control, smooth, self lubricating, can be stopped under load, overload protection by relief valve.

Limitations

- Oil **leakage and mess** , fire hazard, needs pump, cooling and filtering, expensive, noisy, viscosity changes with temperature, contamination and

maintenance, slower than electric / pneumatic.

5. Pneumatic Actuation Systems

- Use **compressed air** (4 to 10 bar, normally 6 bar) to drive actuators. Air is available free, clean, safe (no fire or spark) and the exhaust goes to atmosphere (no return line).
- Used for fast on/off motions - clamping, pick and place, packaging, door opening, tools (drill, hammer), automation.

Components

- **Compressor:** piston/screw/vane compressors, **air receiver (tank)** , after cooler and dryer.
- **FRL unit (Filter, Regulator, Lubricator)** - service unit that cleans, controls pressure and oils the air.
- **Control valves:** DCV (3/2, 5/2, 5/3), **flow control** (speed control by throttling), **non return, quick exhaust, shuttle (OR) and two pressure (AND)** valves.
- **Actuators: pneumatic cylinders** (single acting, double acting, rodless, rotary), air motors.

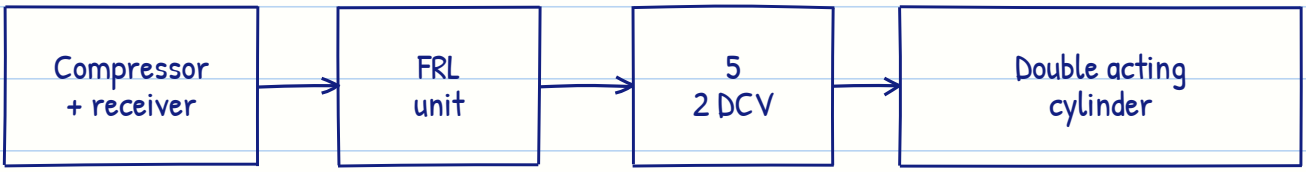


Fig: Basic pneumatic system

Pneumatic vs Hydraulic

Point	Hydraulic	Pneumatic
Medium	Oil	Compressed air
Pressure	Up to 350 bar	4 to 10 bar
Force	Very high	Low to medium
Speed	Slow, smooth	Fast, jerky
Compressibility	Incompressible, stiff	Compressible, spongy
Leakage	Messy oil	Harmless air
Cost	High	Low
Return line	Required	Not needed (exhaust)

Limitations of pneumatic

- Compressible air gives poor position control and low stiffness; limited force; noise of exhaust; moisture in air; energy loss in air generation.

6. Design of Hydraulic and Pneumatic Circuits

- Circuits are drawn using **ISO 1219 graphical symbols** . Draw supply at

the bottom, control at the middle, actuator at top, cylinder on the right. Label components (1.0 cylinder, 1.1 valve, 0.1 supply).

Steps of circuit design

- Understand the task (load, stroke, speed, sequence); select cylinder size from F = P A (add 10-25% for friction); choose valve type and control method (manual, solenoid, pilot); add speed control and safety (relief, check valve); select pump / compressor, tubing, filter; draw the circuit and test.

6.1 Control of a single acting cylinder

- A **3/2 DCV** is used. Pressed: air goes to cylinder, it extends. Released: spring returns the valve, air exhausts, the cylinder spring returns the piston.

6.2 Control of a double acting cylinder

- A **4/2 or 5/2 DCV** (hydraulic: 4/3 valve) - piston extends with one position and retracts with the other. Speed control by **flow control valves** in meter-in or meter-out. Meter-out is preferred for stable speed.

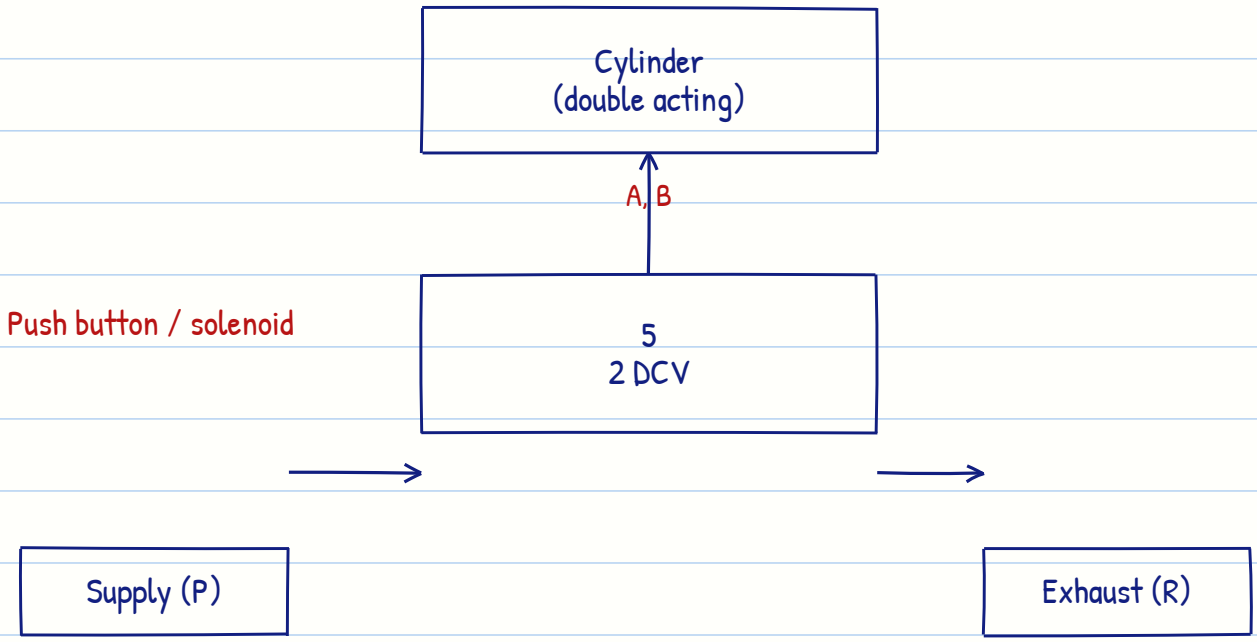


Fig: Double acting cylinder control with 5/2 valve

6.3 Speed control and special circuits

- **Meter-in:** throttle in supply line; **meter-out:** throttle in exhaust line; **bleed-off:** part of oil goes back to tank (hydraulic).
- **Regenerative circuit:** oil from the rod side is joined with the pump flow - fast extension. **Sequencing / counter balance valve** keeps a vertical load from falling. **Pressure reducing valve** for a lower clamp pressure.
- **AND (two pressure) valve:** safety with two hands. **OR (shuttle) valve:** operate from either of two places.

6.4 Sequence circuit (A+ B+ B- A-)

- Two cylinders A and B: A extends, then B extends, B retracts, and A retracts. **Limit switches / roller valves** at the ends give the signal for the next step; use **cascade or electro pneumatic (relay / PLC)** control to avoid overlapping signals.

6.5 Hydraulic circuit example

- A press circuit: pump -> relief valve -> 4/3 DCV (tandem centre, unloads

pump) -> double acting cylinder, with a counterbalance valve and a pressure gauge.

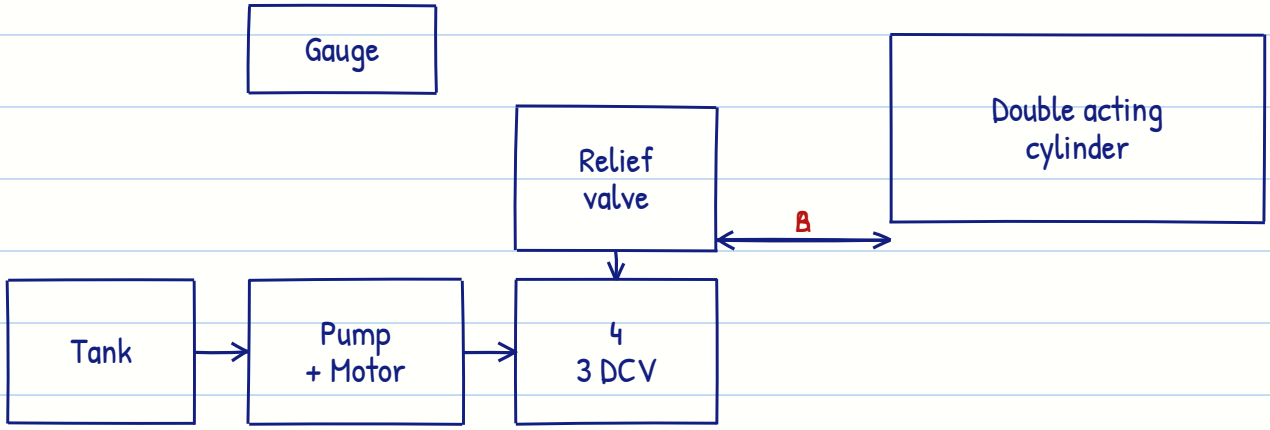


Fig: Hydraulic circuit with 4/3 DCV

7. Piezoelectric Actuators

- **Piezoelectric effect:** certain crystals (quartz, **PZT – lead zirconate titanate** , PVDF) develop an electric charge on mechanical stress (direct effect – sensor). The **converse effect:** they change shape when voltage is applied – used as actuator.
- A stack of thin PZT layers (multilayer stack) gives larger displacement. Strain is about 0.1% – so displacement is only a few microns to tens of microns; **bender and amplified** types give more stroke.
- Displacement dL = d33 x V (for stack with n layers, x n). Response time in microseconds, force up to kN.

Characteristics

- **Very high resolution (nm), fast response (kHz), high force, no**

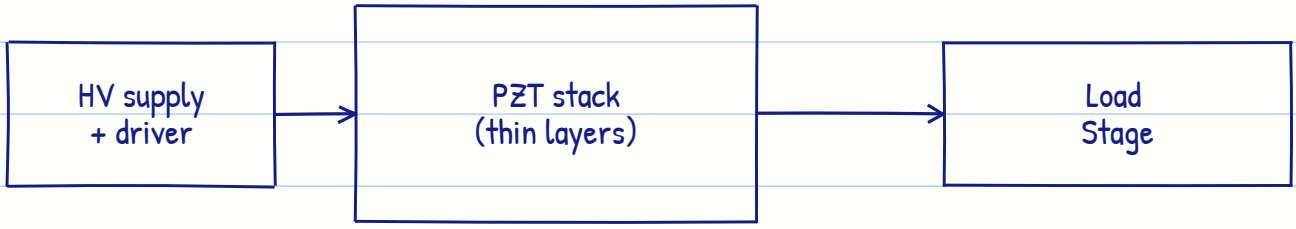
magnetic field, no backlash, and low power at hold. Works in vacuum / clean room; high stiffness.

Limitations

- **Small stroke,** needs **high voltage** (100-1000 V) drive, hysteresis and creep, brittle ceramic, cannot take tension, cost, temperature limit (Curie point).

Applications

- Inkjet printer heads, fuel injectors (diesel/gasoline), nano positioning stages, atomic force microscope (AFM), ultrasonic motors and cleaners, autofocus, vibration damping, micro pumps, MEMS.



V up -> length increases (micro displacement)

Fig: Piezoelectric stack actuator

8. Shape Memory Alloys (SMA)

- SMA are metals which **remember their original shape** . After deformation at low temperature they recover the shape on **heating** above the transformation temperature. Eg: **Nitinol (Nickel-Titanium, 55% Ni - 45% Ti)** , Cu-Zn-Al, Cu-Al-Ni.
- **Principle:** solid state phase change between **martensite** (low temperature, soft, easily deformed) and **austenite** (high temperature,

stiff, original shape). Transformation temperatures: Ms, Mf (martensite start / finish) and As, Af (austenite start / finish).

- **One way memory effect:** deform in martensite -> heat -> recovers the shape; on cooling it does not change shape back until deformed again. **Two way:** remembers both hot and cold shapes. **Superelasticity (pseudo elasticity):** recovers large strain (8%) on unloading at constant temperature.
- **Heating** is done by passing an electric current through the wire (Joule / resistance heating) - so it acts as an electrical actuator. Cooling by air.

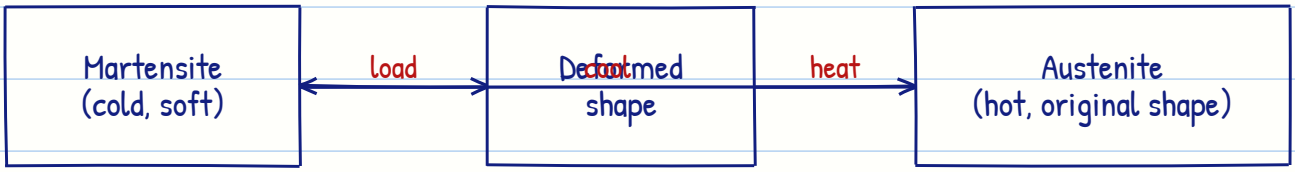


Fig: Shape memory effect cycle

Characteristics

- Recovery strain up to 4-8%, **high force per unit weight** , silent, compact, simple, biocompatible, corrosion resistant.

Limitations

- **Slow** (cooling is slow, 1-few Hz), low efficiency (about 5%), hysteresis, fatigue (limited life), complex control, expensive, small strokes.

Applications

- Robot grippers and artificial muscles, micro actuators, valves, thermostat,

couplings, aerospace (morphing wing), medical stents and guidewires, dental braces, eyeglass frames, fire safety.

9. Other Smart Actuators (brief)

- **Magnetostrictive actuators** (Terfenol-D): change length in a magnetic field; high force, fast; used in sonar and precision positioning.
- **Electro / Magneto-rheological fluids:** viscosity changes under field - dampers, clutches, brakes.
- **Electro active polymers (EAP), IPMC:** artificial muscles; **electrostrictive** and **thermal bimorph** actuators in MEMS.
- **Electrostatic (comb drive) actuators** in MEMS - covered in Unit V.

10. Selection Criteria for Actuators

- **Load requirement:** force or torque, speed, stroke / angle, power, duty cycle and load inertia (match inertia ratio for motors).
- **Accuracy, resolution and repeatability** - stepper for light open loop, servo for high accuracy, piezo for nanometre.
- **Speed and response (bandwidth):** pneumatic fast on/off, hydraulic smooth, servo fast.
- **Power source available:** electrical supply, compressed air or hydraulic power pack.
- **Controllability and compatibility** with the controller (PWM, pulse, voltage, current signal).
- **Size, weight, and mounting;** environment (explosive, dust, temperature, wash down).
- **Efficiency, power consumption, noise, vibration;** safety.

- **Reliability, life and maintenance; cost** (initial + running) and availability.

Actuator	Force / torque	Accuracy	Typical use
Pneumatic	Low-medium	Low (on/off)	Pick & place, clamping
Hydraulic	Very high	Medium	Press, excavator
Stepper	Low-medium	Good (open loop)	3D printer, small CNC
Servo	Medium-high	Very high	CNC, robots
Piezo	High but tiny stroke	nm level	Nano stage, injector
SMA	High force / weight	Low, slow	Micro gripper, valve

Important Questions (Unit III)

- Explain mechanical, electrical, hydraulic and pneumatic actuation systems with their limitations.
- Compare stepper and servo motors. Explain the working of a stepper motor.
- Compare hydraulic and pneumatic systems. Explain the components of a hydraulic system.
- Design a pneumatic circuit for a double acting cylinder / sequence A+ B+ B- A-.
- Explain piezoelectric actuators and shape memory alloys with applications.
- What are the selection criteria for actuators?